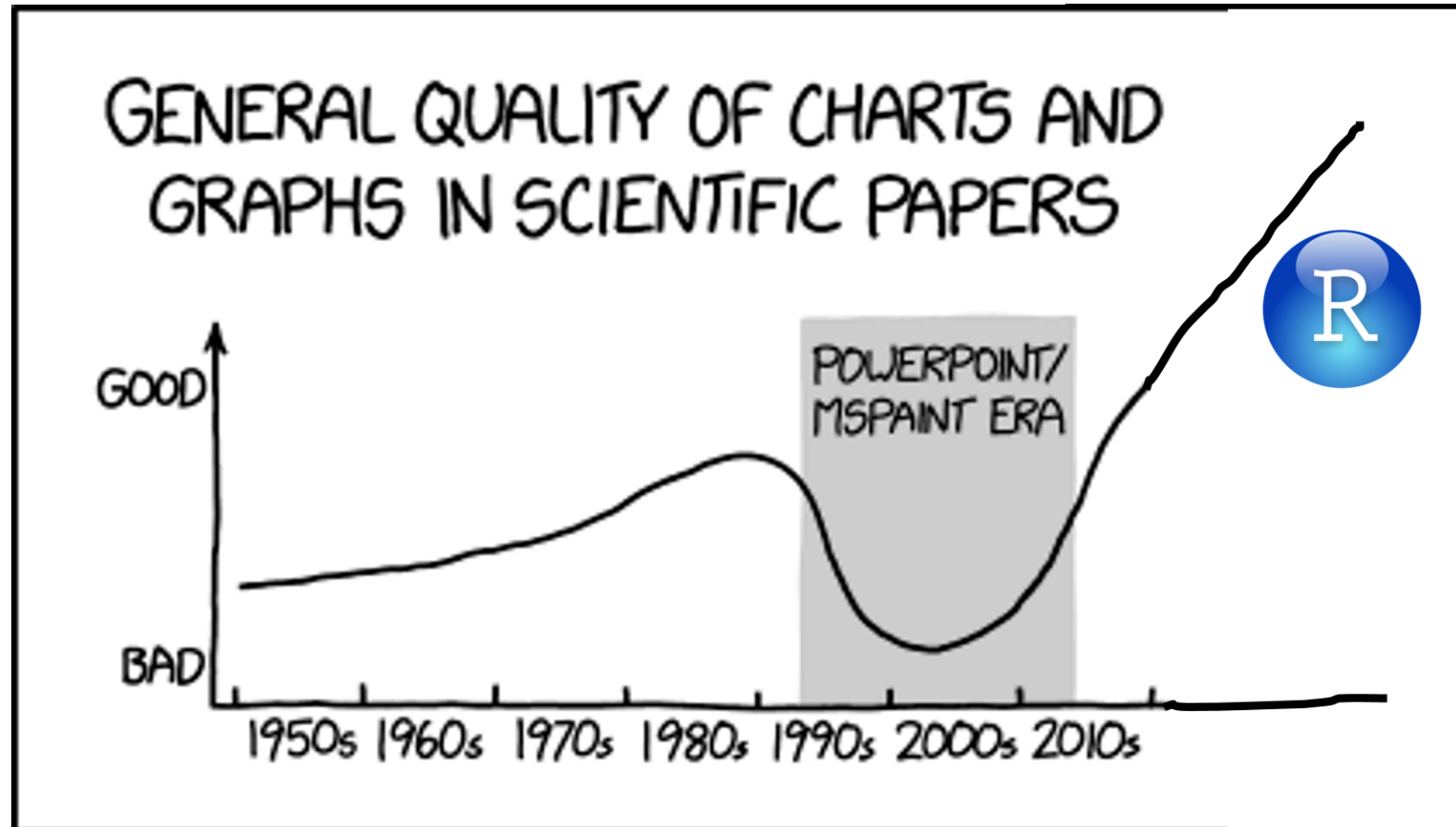


DS 3100

Fundamentals of Data Science

Exploratory Data Analysis



Learning Goals

- Understand what EDA can and can't do
- Develop an intuition of plot types with data types
- Create fully customizable graphics using `{ggplot2}`
- Use graphics to
 - summarize characteristics of the data
 - answer research questions
 - communicate clearly and concisely your results

Motivation

- Exploratory Data Analysis (EDA) encompasses analytic methods aimed at determining and describing characteristics of data
 - Contrast with inferential analysis or inferential statistics which aims to use the scientific method to answer research questions
- Examples
 - Which values are the most common? Why?
 - Which values are rare? Why? Does that match your expectations?
 - Can you see any unusual patterns? What might explain them?
- Where does data visualization come in?
 - Data visualization is one of the primary and fundamental tools for EDA
 - It's not the only option for EDA, but it will be the focus of this section

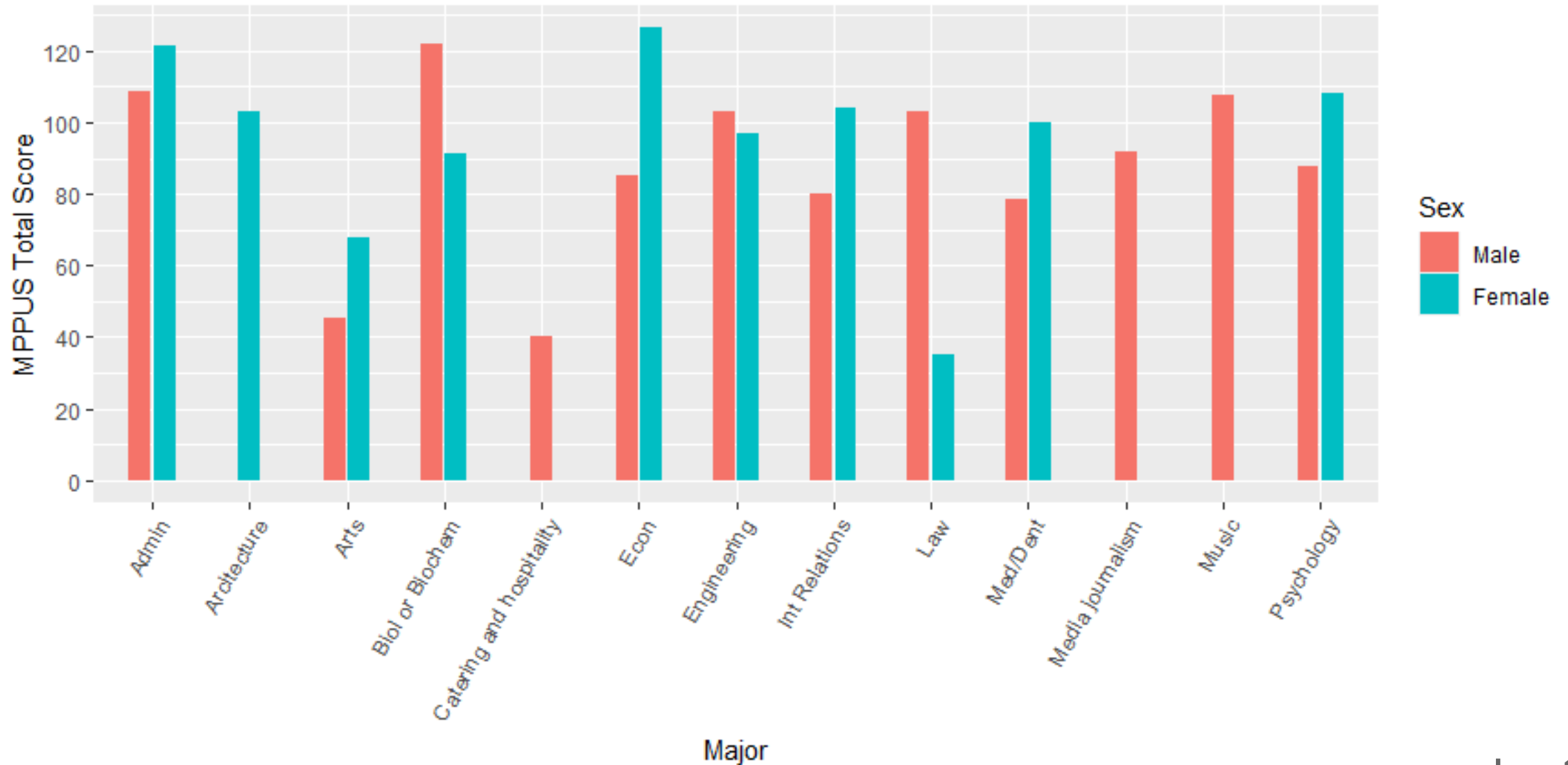
ID	Major_Studied	Age	Sex	MPPUS_Total	GPA	Extraversion	Neuroticism
1	Psychology	21.14990	Female	128	2.93	17	11
2	Psychology	20.99384	Female	183	2.98	15	21
3	Psychology	22.97604	Female	129	3.57	19	5
4	Psychology	22.79535	Female	116	3.98	17	15
5	Psychology	20.00821	Female	67	3.54	20	10
6	Psychology	20.98836	Female	143	3.22	14	16
7	Arts	25.81246	Male	38	4.00	14	21
8	Psychology	27.18686	Male	88	2.90	18	7
9	Law	18.71869	Female	35	3.81	20	13
10	Admin	19.98084	Female	95	3.33	17	14
11	Engineering	22.29706	Male	114	3.33	13	9
12	Engineering	21.84257	Male	117	3.02	18	13
13	Law	20.56674	Male	122	999.00	22	10
14	Engineering	23.30185	Male	82	2.76	17	8
15	Engineering	22.04517	Male	132	2.90	17	9
16	Engineering	21.65914	Female	74	3.14	19	9
17	Engineering	23.15400	Male	148	2.73	20	11
18	Engineering	19.91513	Female	45	3.53	17	2
19	Admin	19.62218	Male	169	3.15	21	10
20	Engineering	23.17591	Male	71	3.24	18	12
21	Engineering	NaN	Male	101	2.37	20	18
22	Engineering	25.37440	Male	106	2.12	21	8
23	Engineering	23.65503	Male	65	3.87	12	20
24	Music	22.24778	Male	100	3.09	19	6
25	Catering and hospitality	23.17591	Male	40	2.97	19	6
26	Psychology	20.49555	Female	61	3.85	19	11

Data

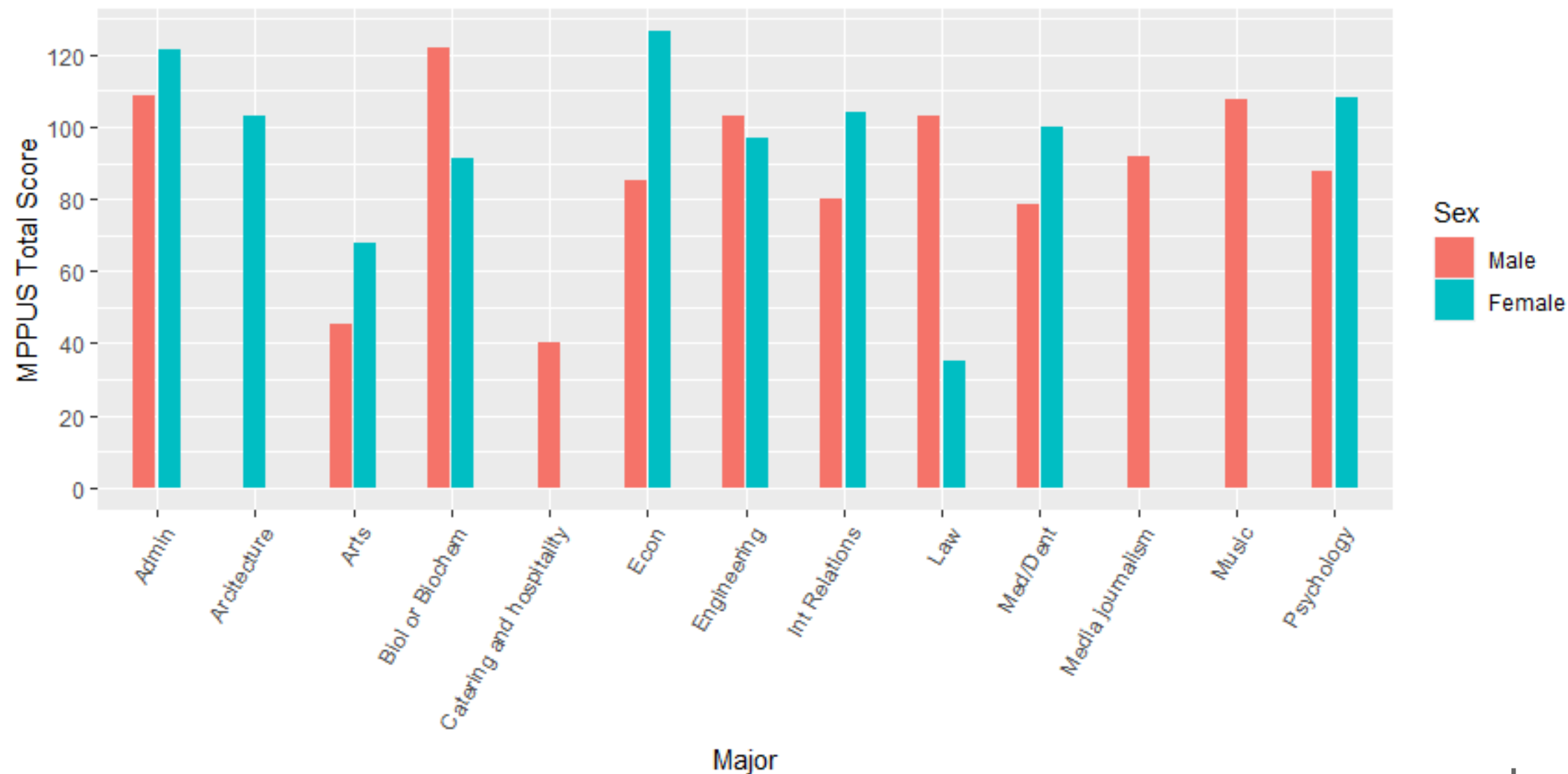
Pluck, G. (2020). Dataset for: Cognitive Ability, Reward Processing and Personality Associated with Different Aspects of Smartphone Use [Data set]. *PsychArchives*.

<https://doi.org/10.23668/PSYCHARCHIVES.2791>

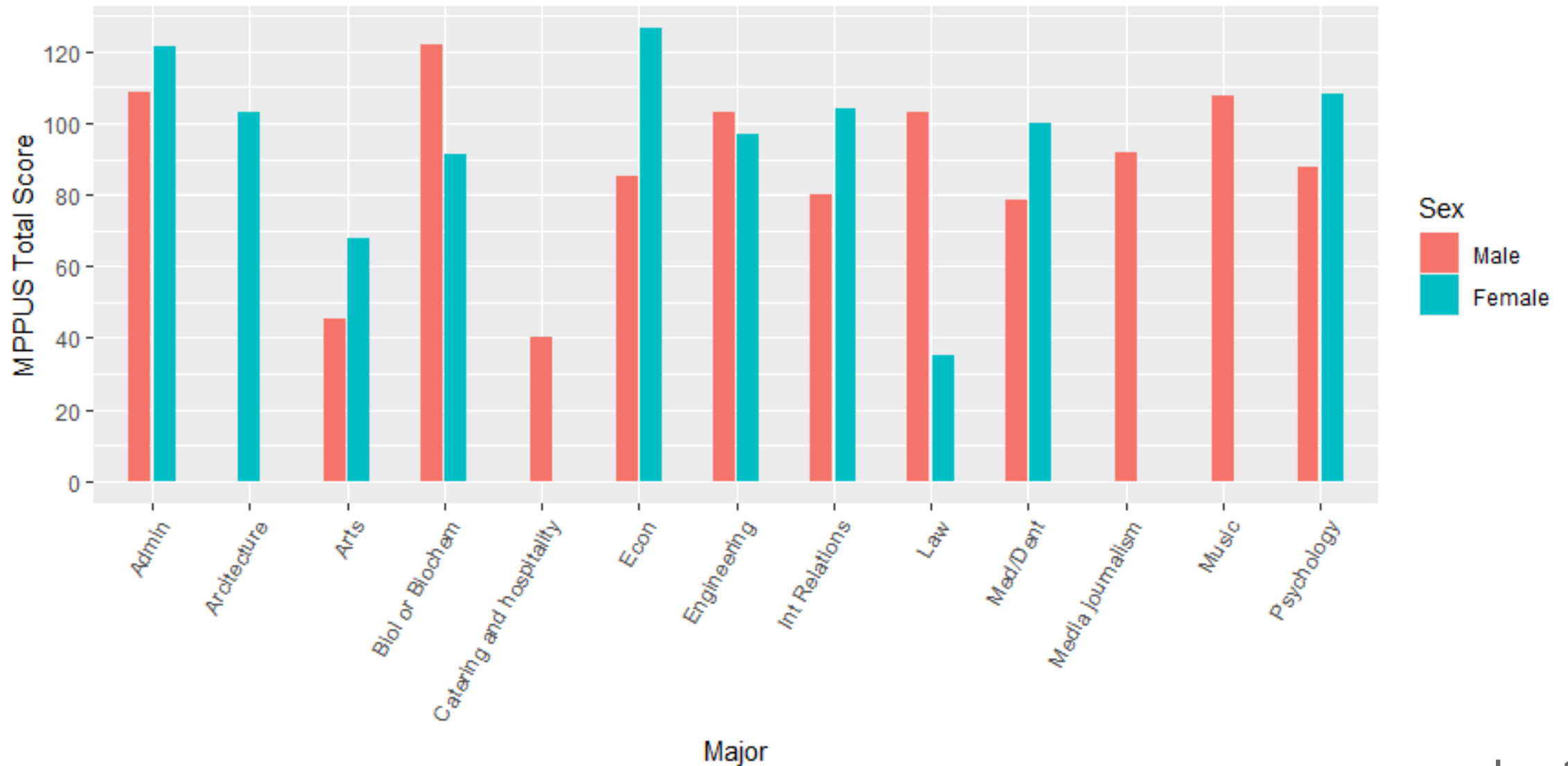
Graphics Organize Data in Communicable Ways



Which major has the largest sex difference in problematic phone usage?



How could this figure be improved?



Figures in R

Base R

- not often used in scientific publications or professional reports
- many times, a quick base R plot (for your own knowledge) is quicker than {ggplot2}
- good place to start exploring

{ggplot2}

- current "gold standard" in data visualization (even compared to Python options)
- complete customization and control over graphics
- **time-intensive**: save efforts until you know *what* and *how* you want to visualize



from Data to Viz

'From Data to Viz' is a classification of chart types based on input data format. It will help you find the perfect chart in three simple steps:

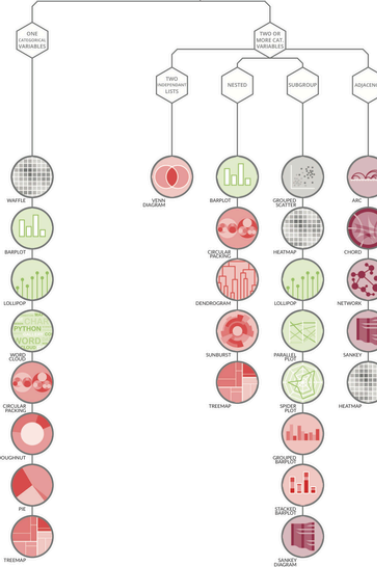
- 1 Identify what type of data you have.
- 2 Go to the corresponding decision tree and follow it down to a set of possible charts.
- 3 Choose the chart from the set that will suit your data and your needs best.

Dataviz is a world with endless possibilities and this project does not claim to be exhaustive. However it should provide you with a good starting point. For an interactive version and much more, visit:

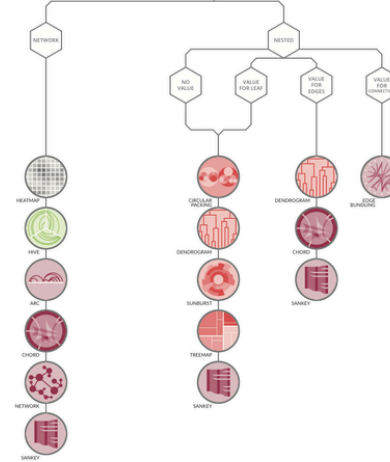
data-to-viz.com

www.data-to-viz.com

CATEGORIC



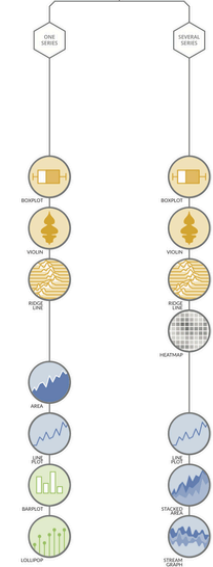
RELATIONAL



MAP



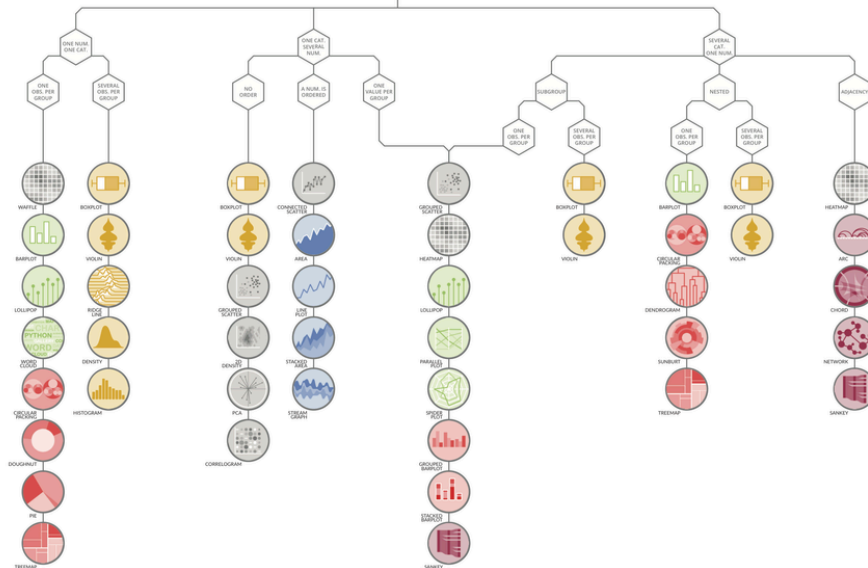
TIME SERIES



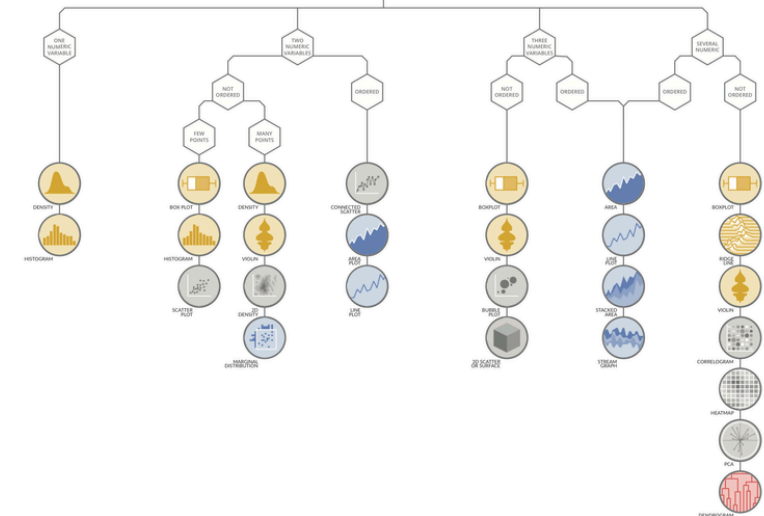
WHAT DO YOU WANT TO SHOW ?

- Distribution
- Correlation
- Ranking
- Part of a whole
- Evolution
- Maps
- Flow

CATEGORIC AND NUMERIC

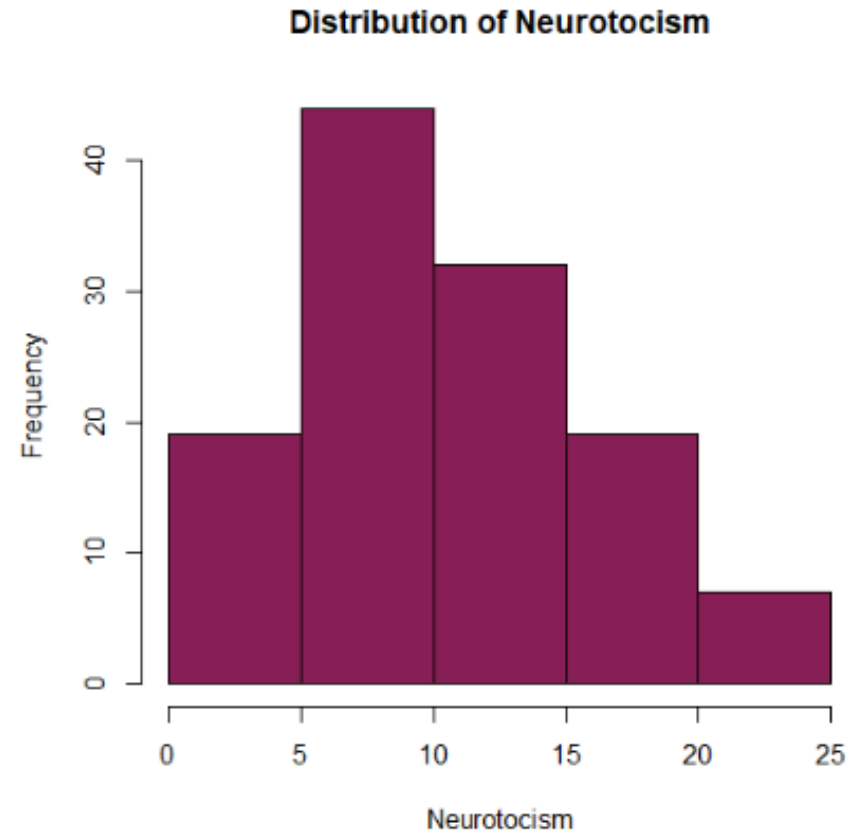
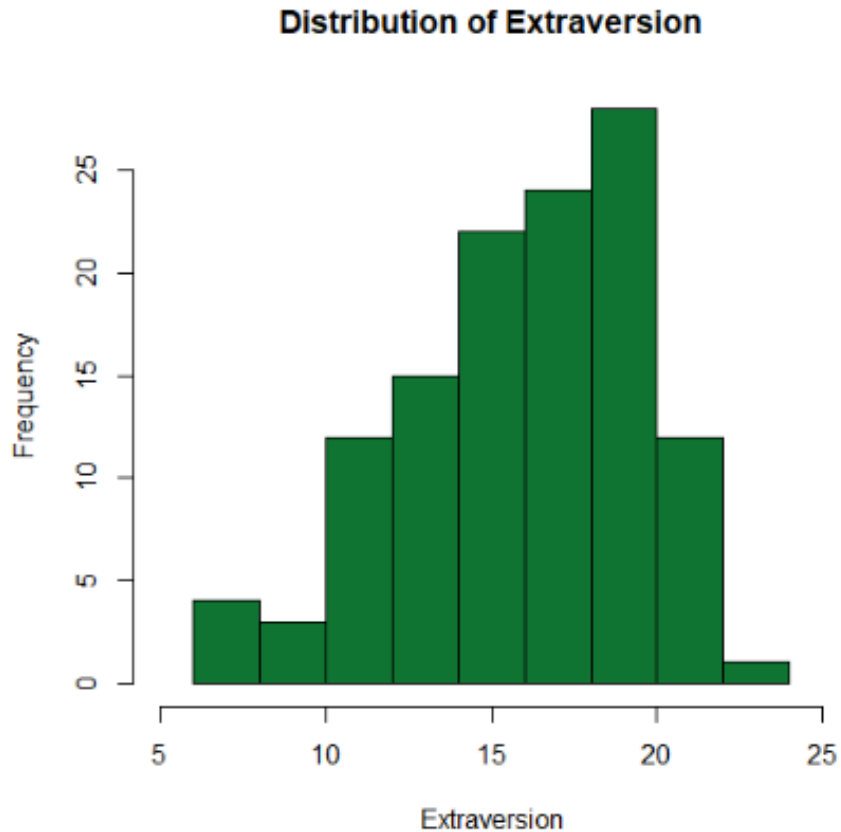


NUMERIC



Univariate Visualization

one variable at a time

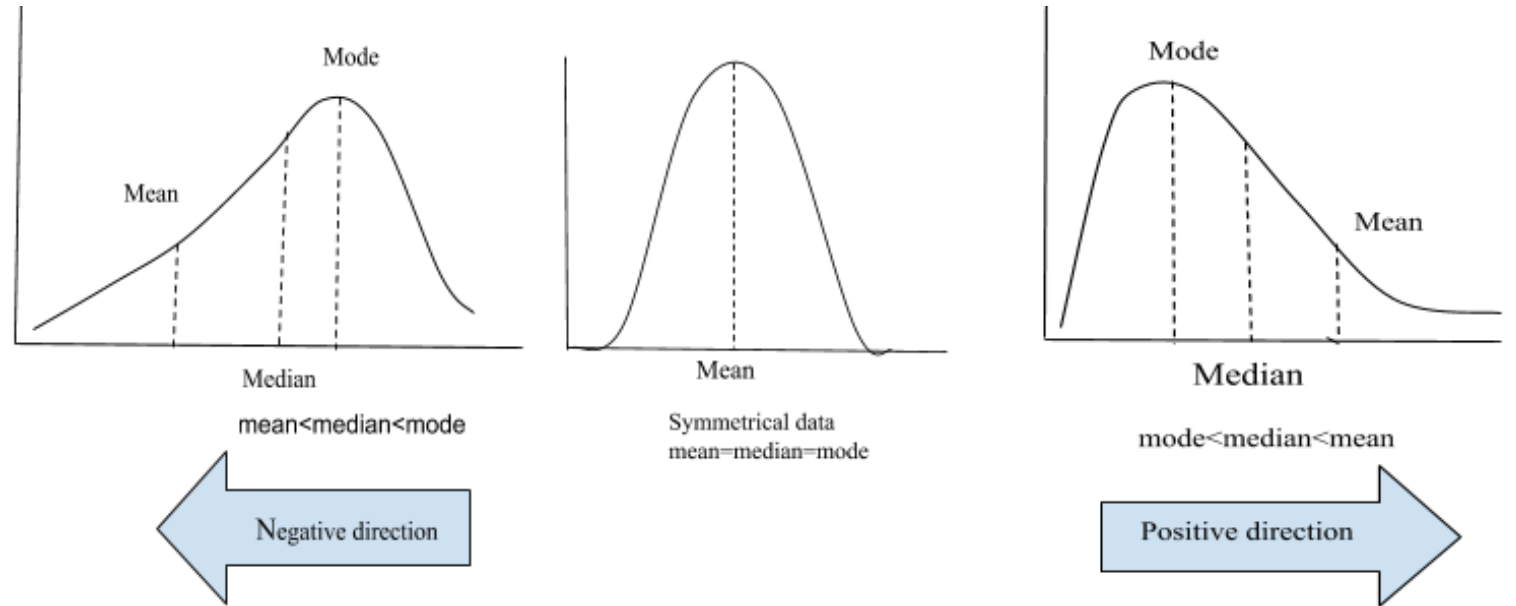


Histogram

Univariate Visualization

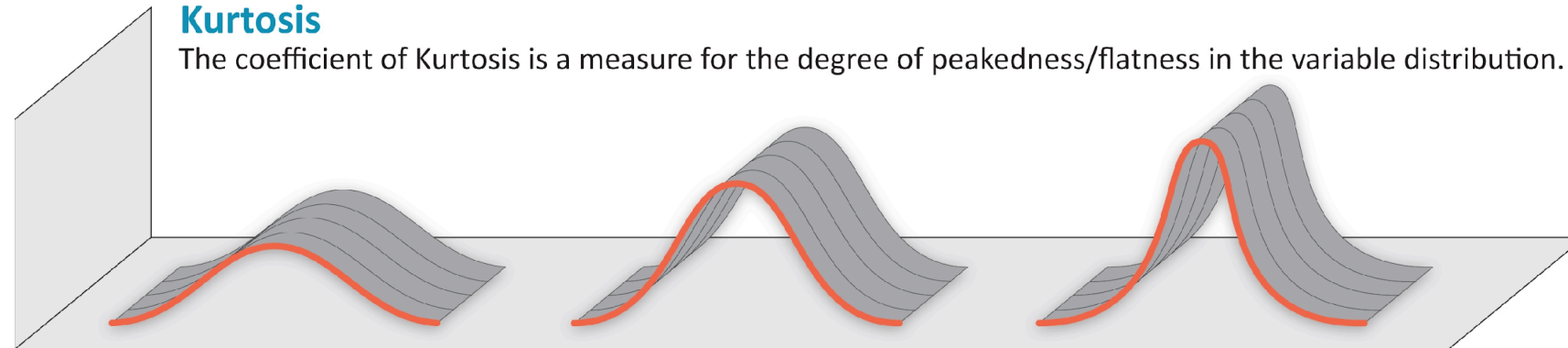
good for properties of distributions

Skewness



Kurtosis

The coefficient of Kurtosis is a measure for the degree of peakedness/flatness in the variable distribution.



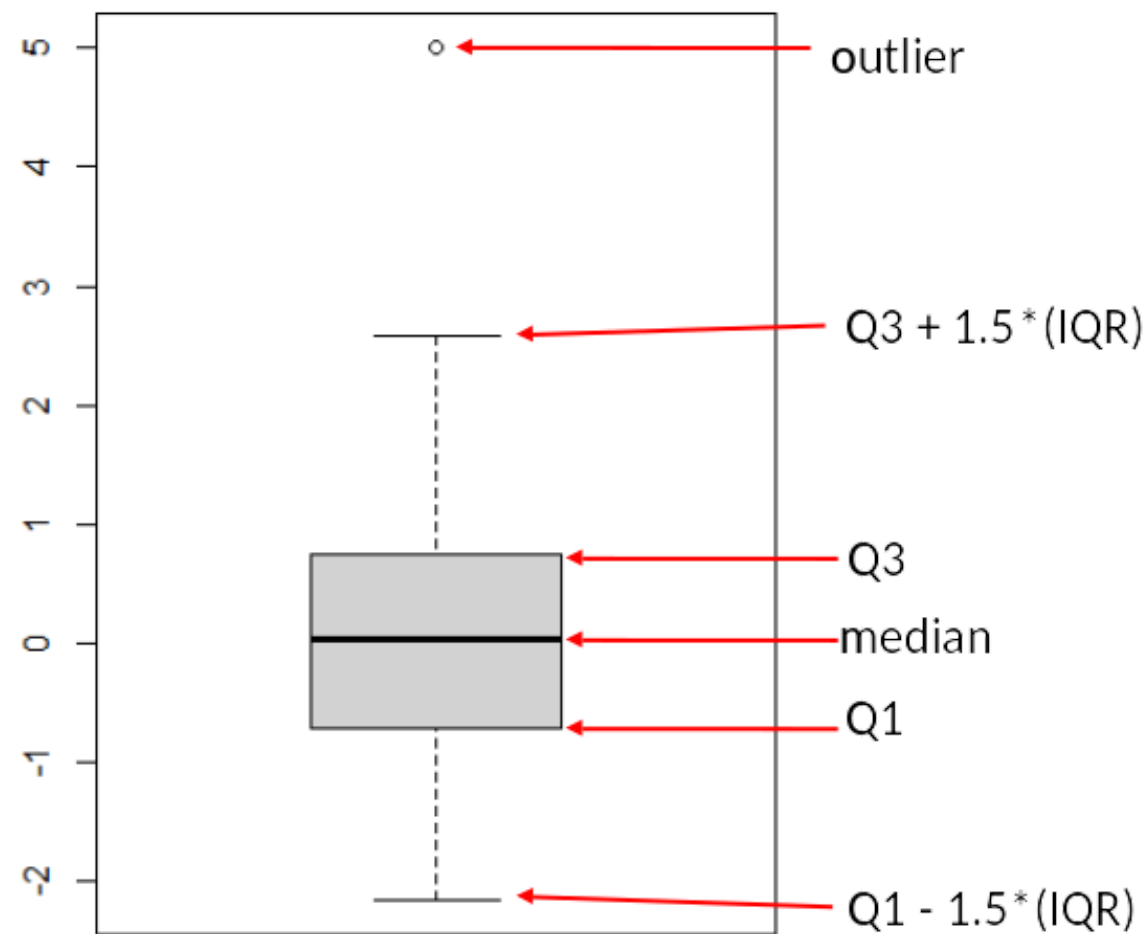
Platykurtic distribution.
Low degree of peakedness.
Kurtosis < 0

Normal distribution.
Mesokurtic distribution.
Kurtosis = 0

Leptokurtic distribution.
High degree of peakedness.
Kurtosis > 0

Univariate Visualization

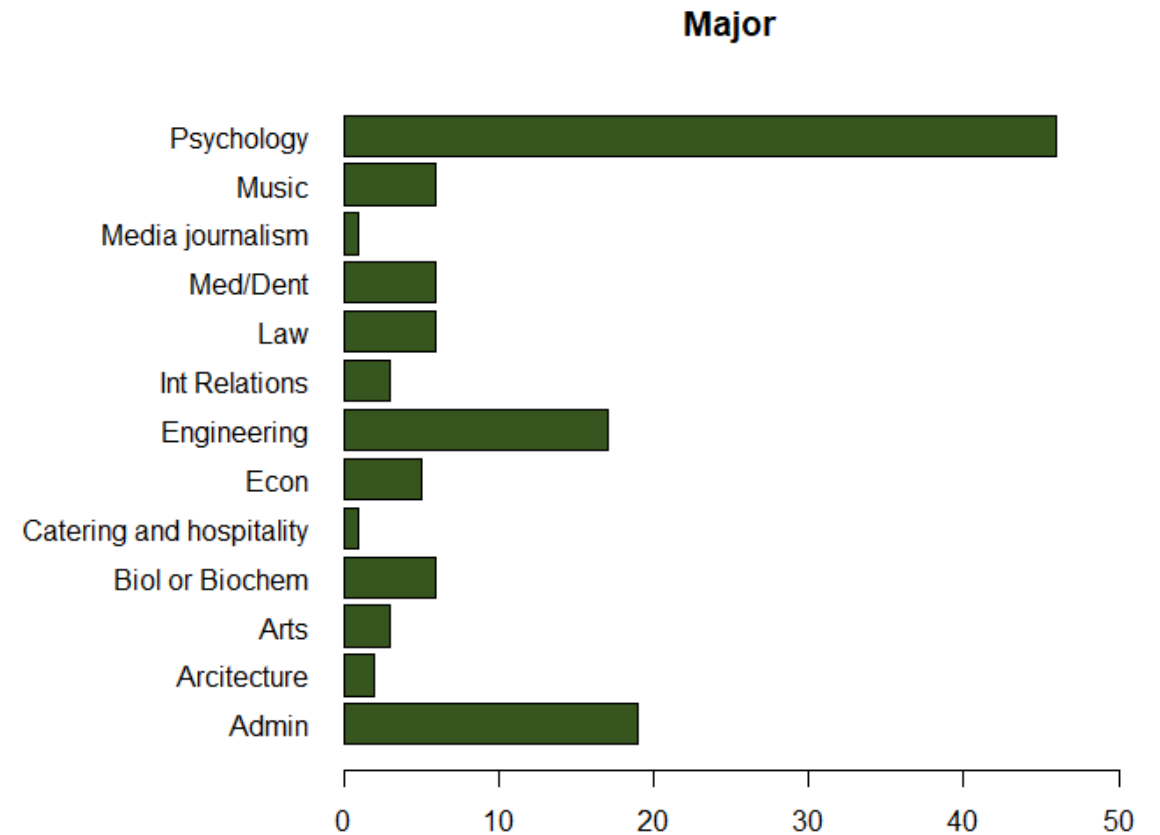
good for comparing distributions



Boxplot

Univariate Visualization

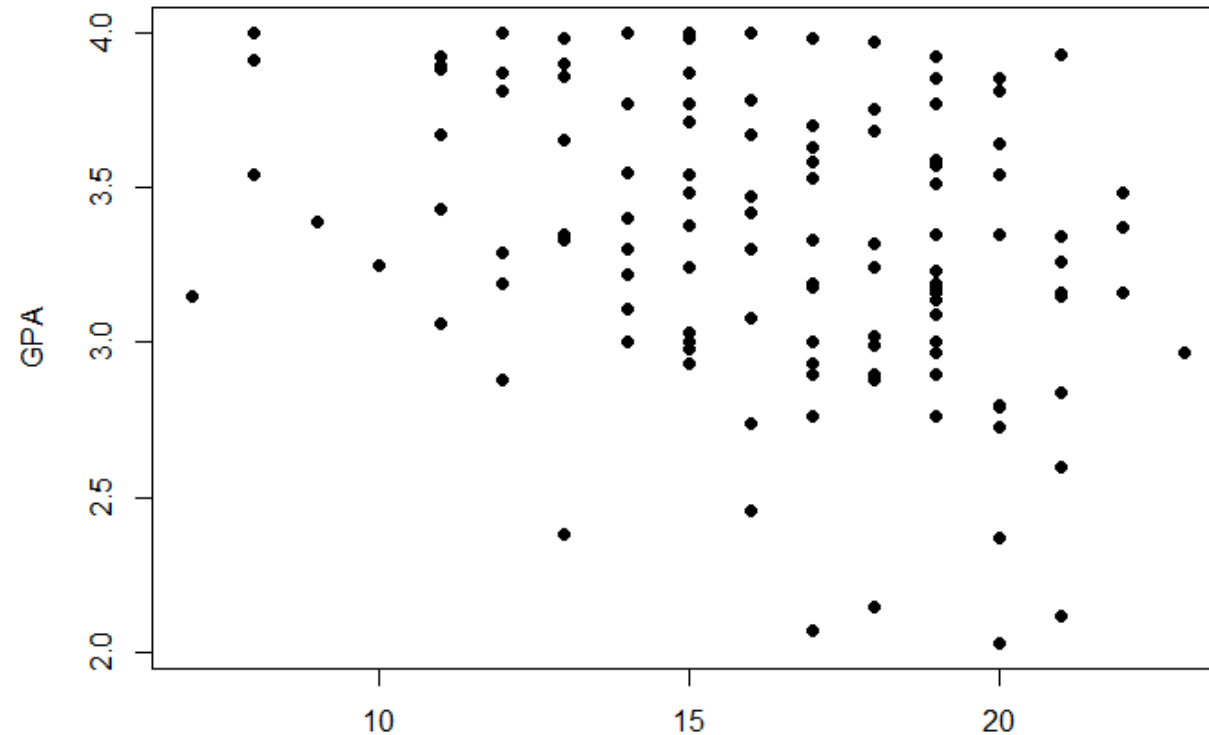
good for comparing counts, frequencies, and proportions



Bar Plot

Bi- and multivariate Visualization

two or more variables at a time

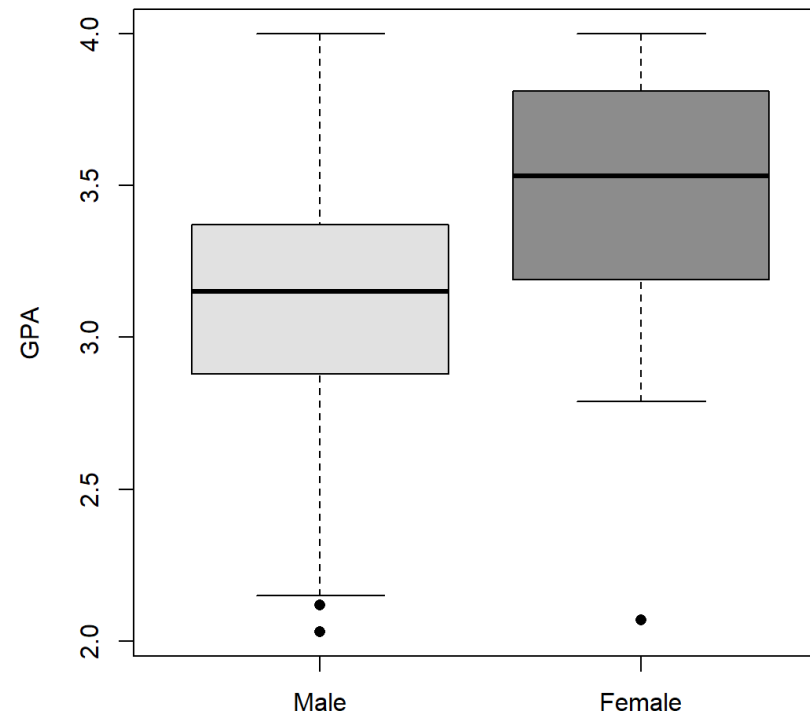


Scatterplot

most useful for visualizing continuous variables

Bi- and multivariate Visualization

two or more variables at a time

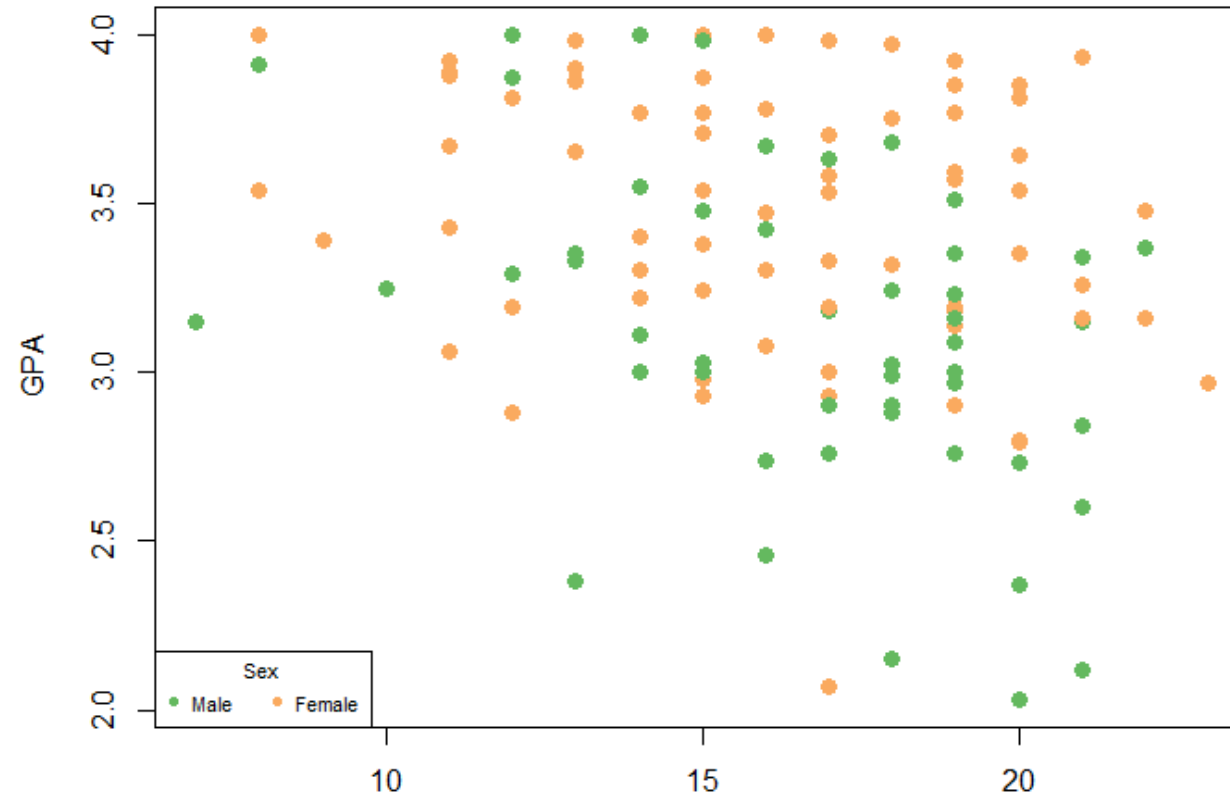


Grouped Boxplot

most useful for visualizing continuous variables separated by group(s)

Bi- and multivariate Visualization

two or more variables at a time

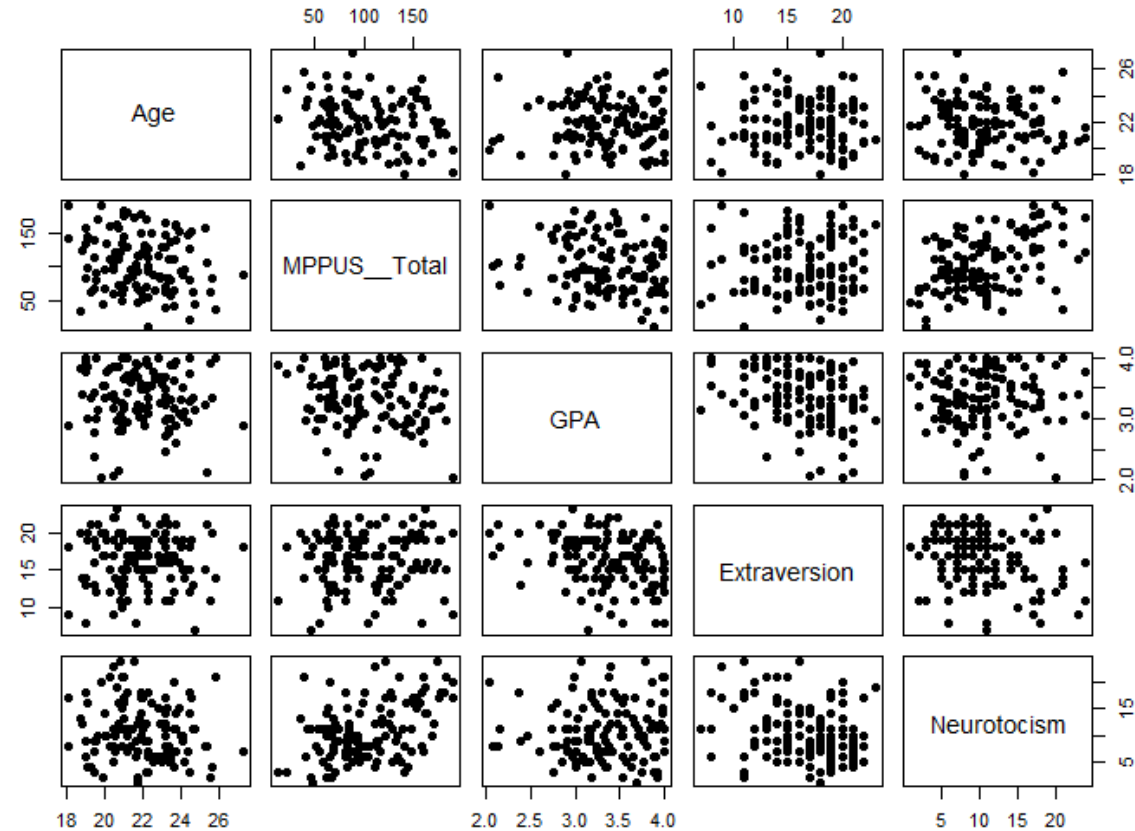


Grouped Scatterplot

most useful for grouped relationships between continuous variables

Bi- and multivariate Visualization

two or more variables at a time

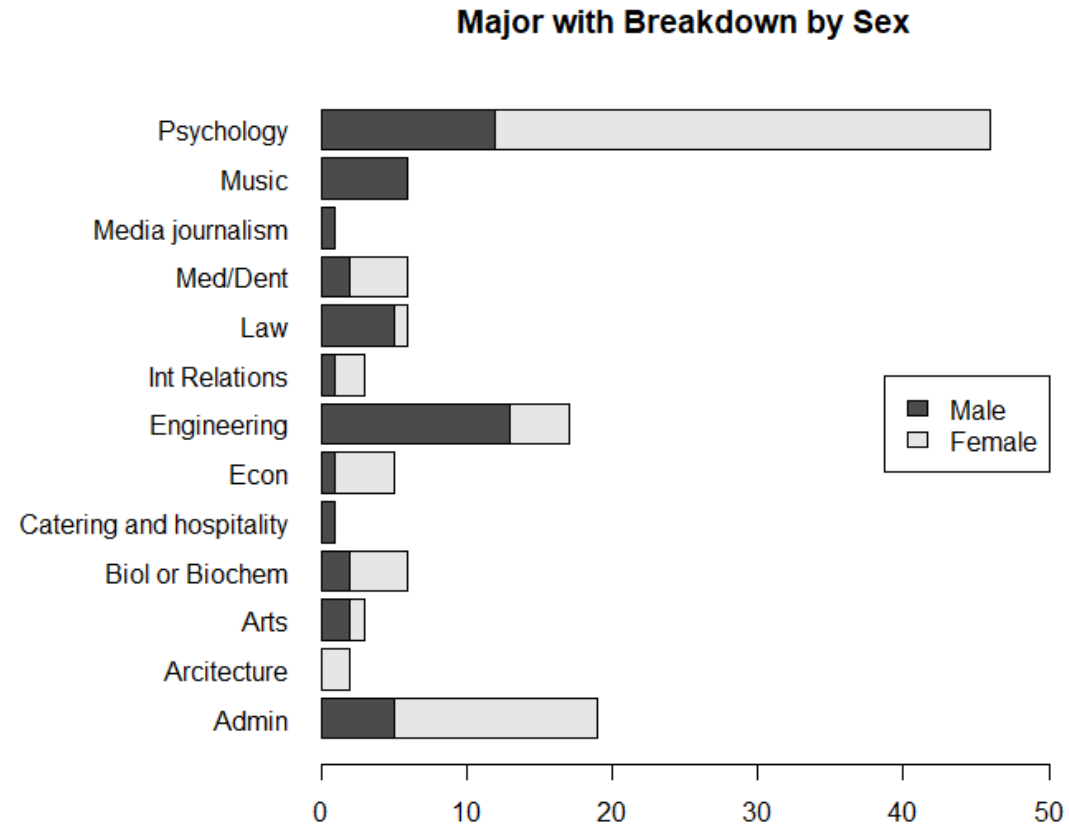


Scatterplot Matrix

most useful for multiple continuous variables

Bi- and multivariate Visualization

two or more variables at a time



Scatterplot Matrix

most useful for multiple categorical variables



from Data to Viz

'From Data to Viz' is a classification of chart types based on input data format. It will help you find the perfect chart in three simple steps:

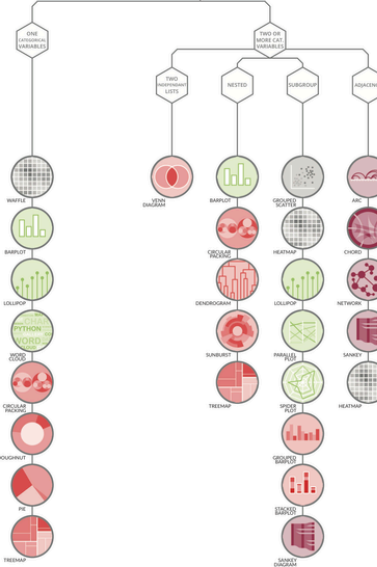
- 1 Identify what type of data you have.
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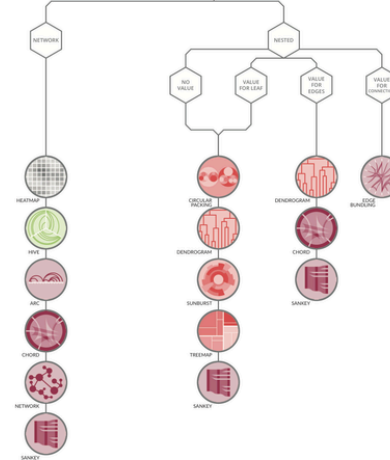
data-to-viz.com

www.data-to-viz.com

CATEGORIC



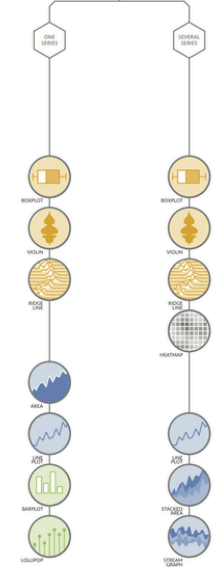
RELATIONAL



MAP



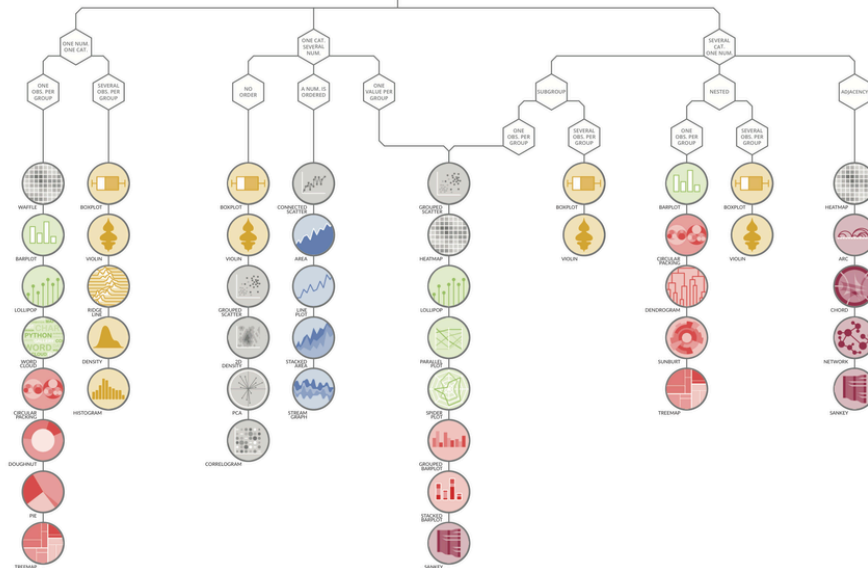
TIME SERIES



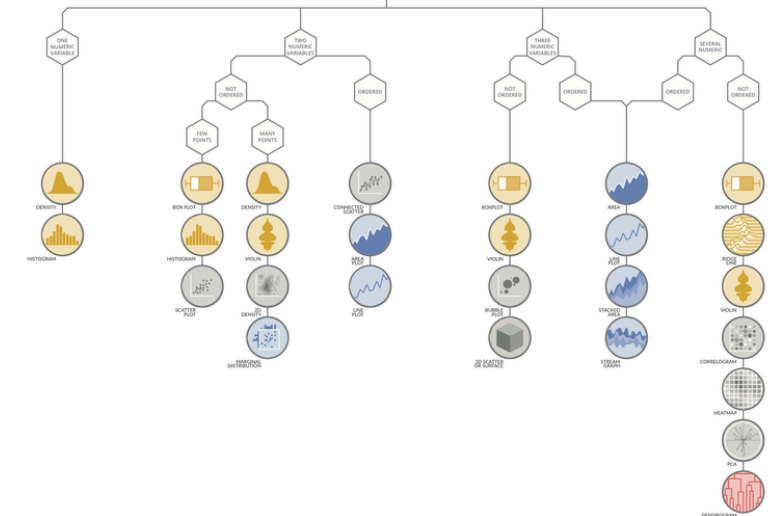
WHAT DO YOU WANT TO SHOW ?

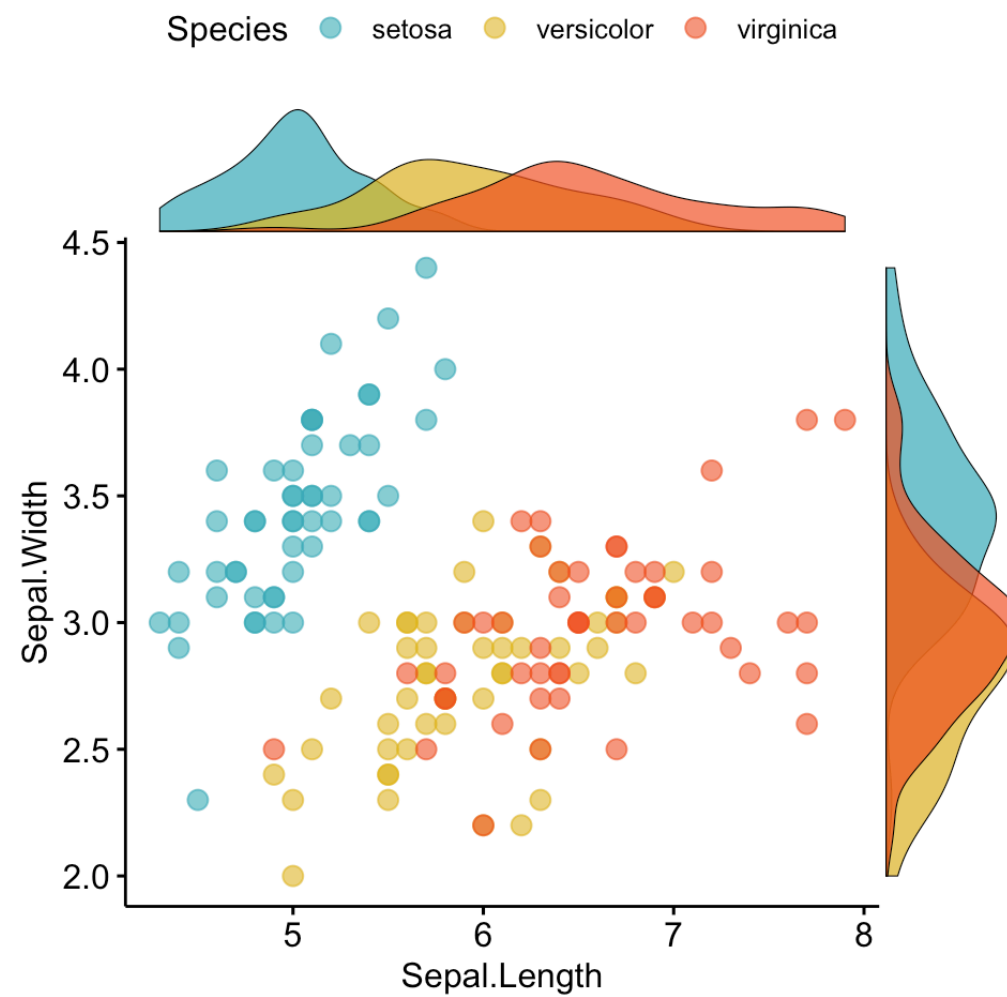
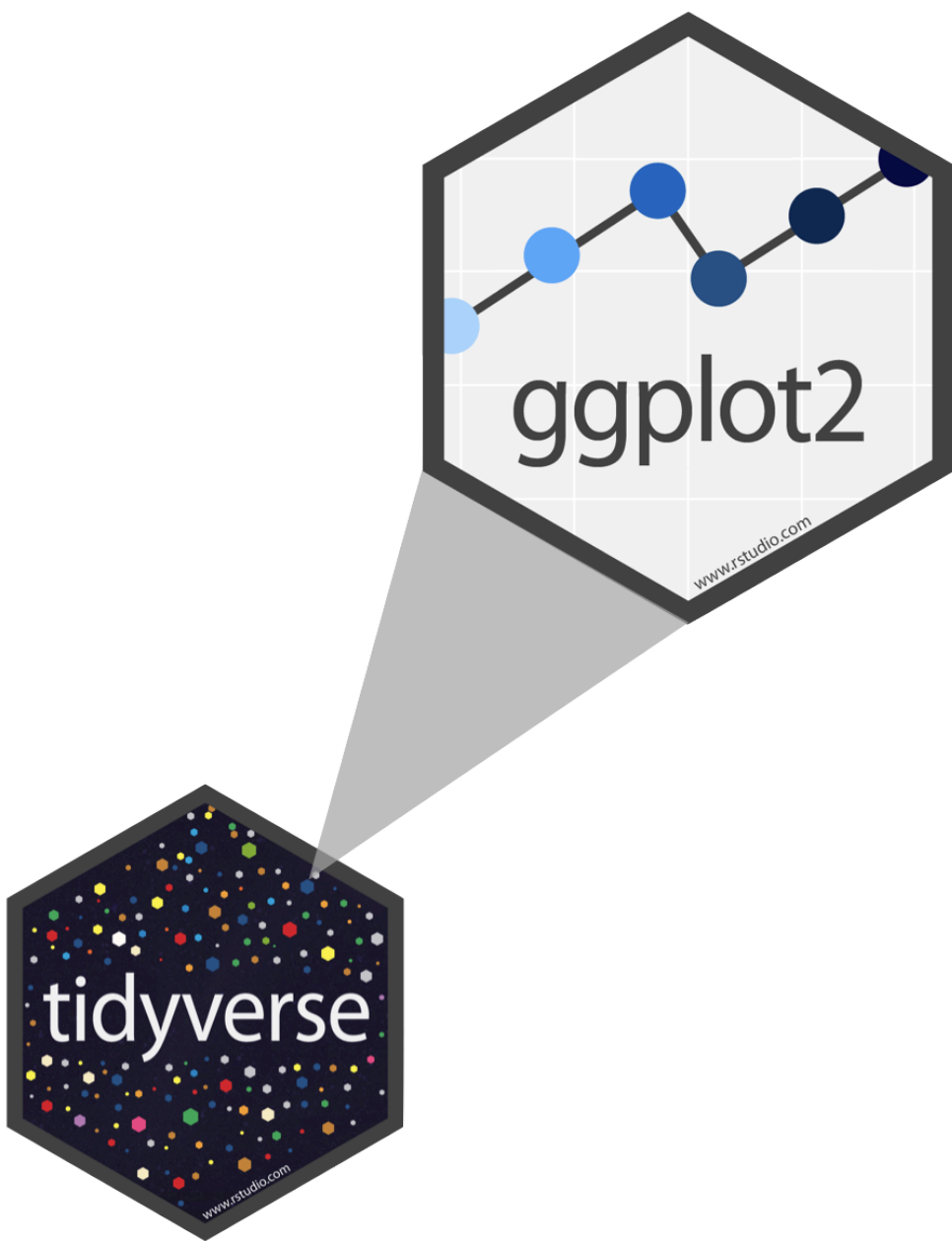
- Distribution
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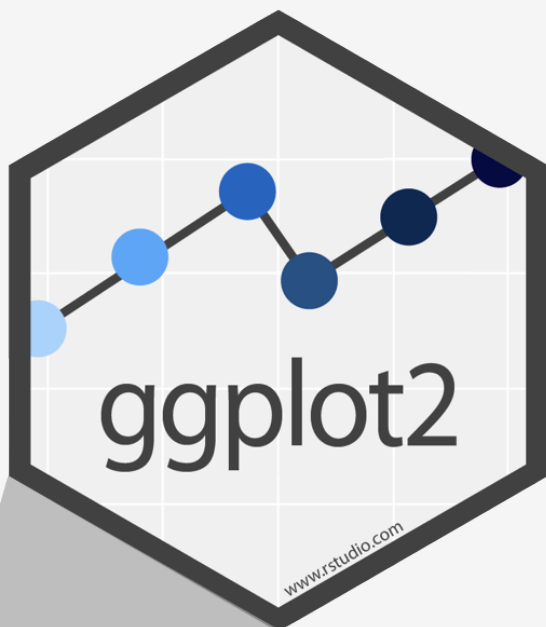
CATEGORIC AND NUMERIC



NUMERIC

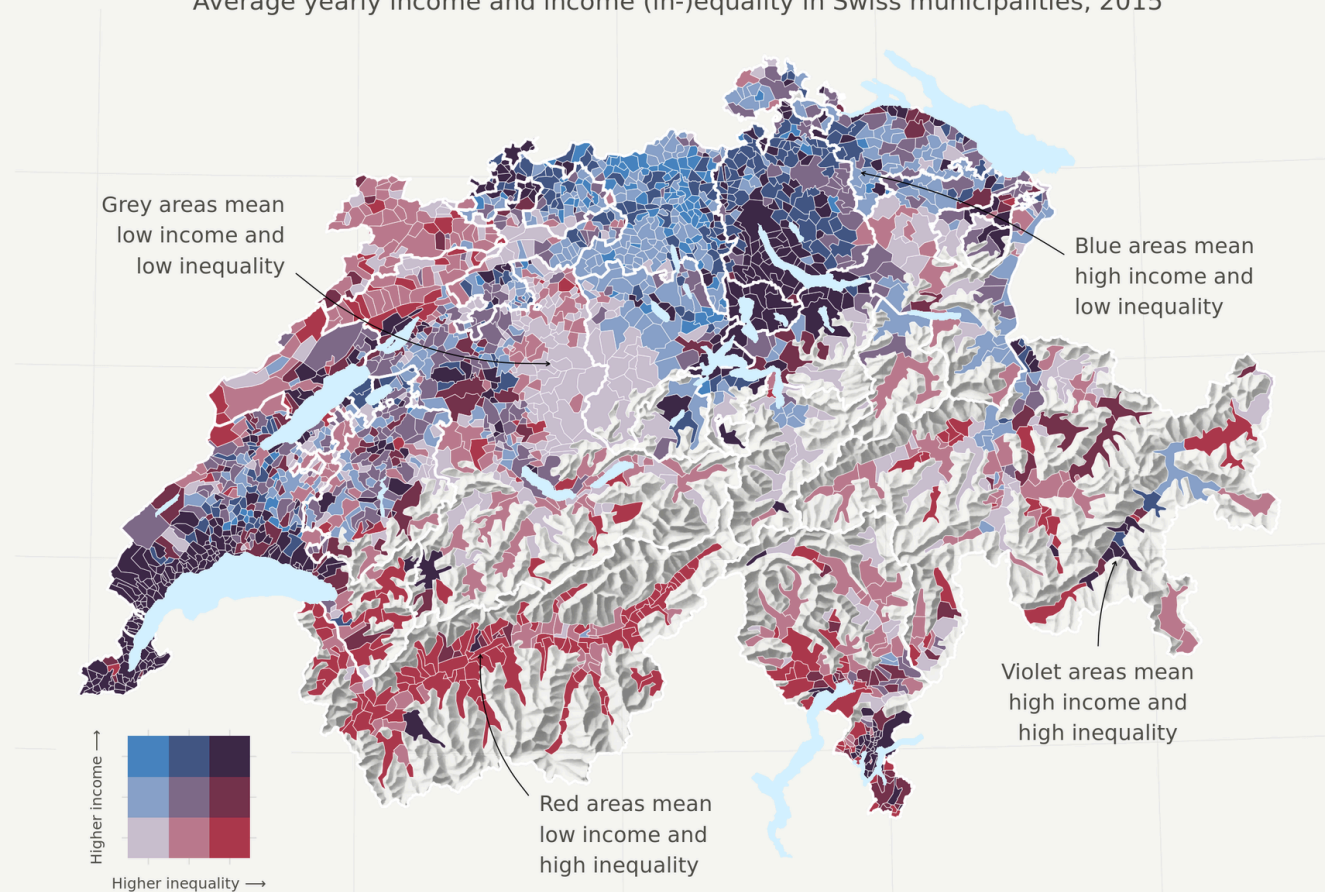




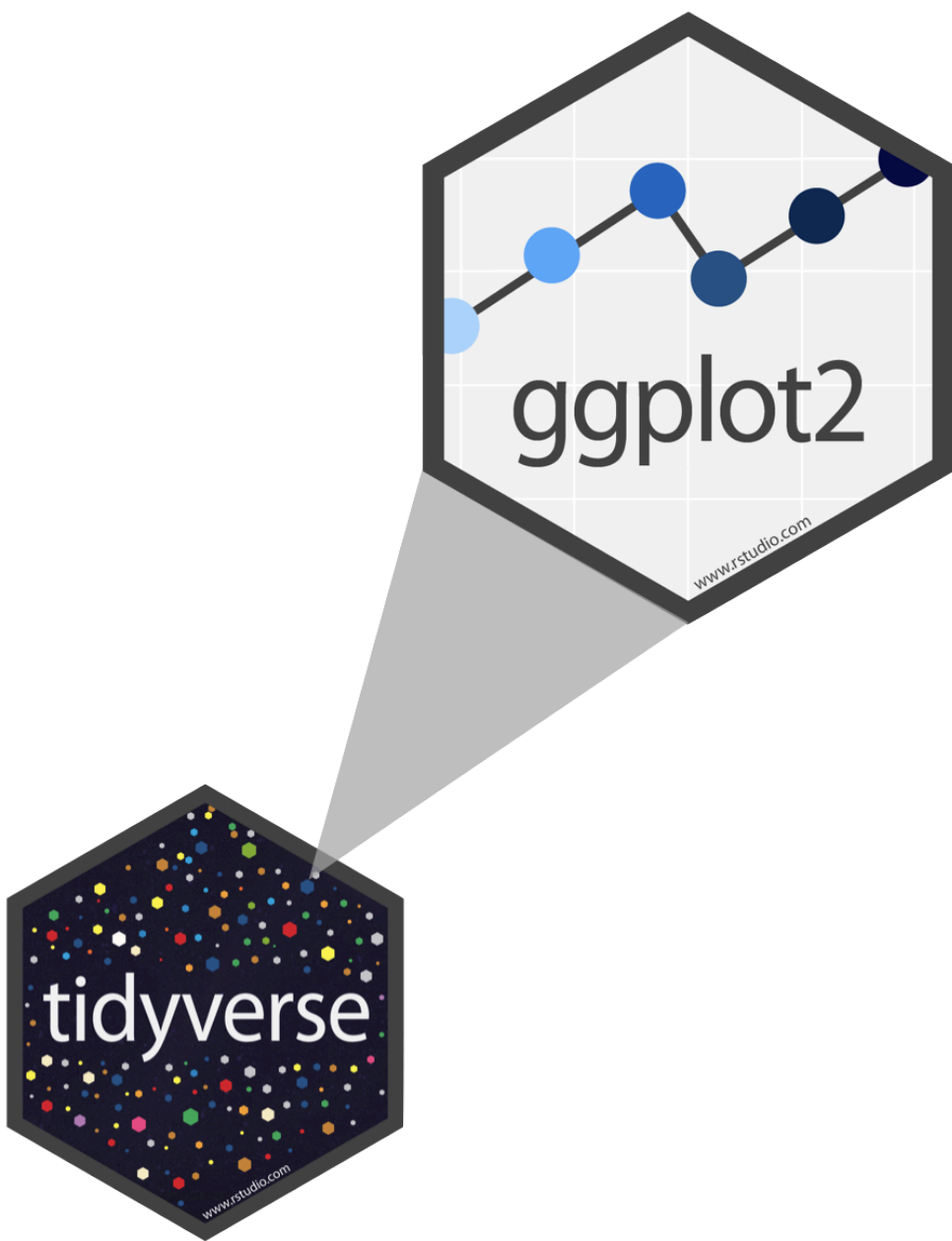


Switzerland's regional income (in-)equality

Average yearly income and income (in-)equality in Swiss municipalities, 2015



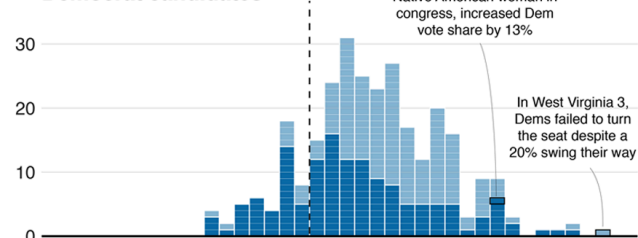
Map CC-BY-SA; Code: github.com/grssnbchr/bivariate-maps-ggplot2-sf
Authors: Timo Grossenbacher (@grssnbchr), Angelo Zehr (@angelozehr)
Geometries: ThemaKart BFS and swisstopo; Data: ESTV, 2015



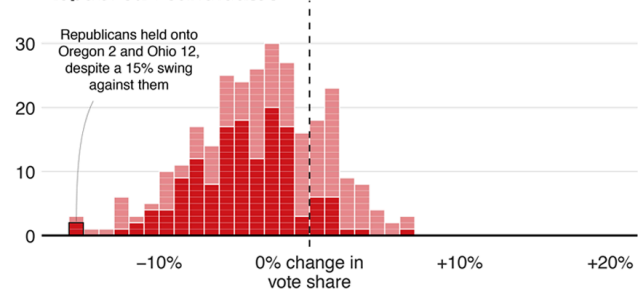
Blue wave

■ Won seat ■ Didn't win

Democrat candidates



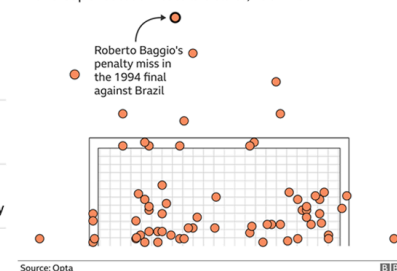
Republican candidates



Source: AP, 19:01 ET

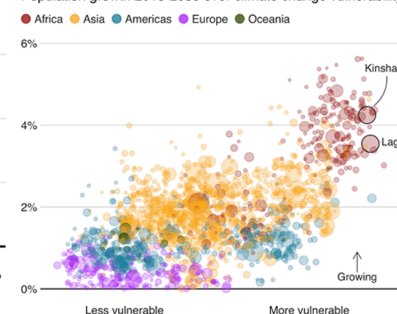
Where penalties are saved

World Cup shootout misses and saves, 1982-2014



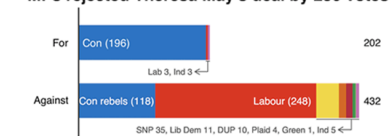
Fast-growing cities face worse climate risks

Population growth 2018-2035 over climate change vulnerability



Source: Verisk Maplecroft. Circle size represents current population.

MPs rejected Theresa May's deal by 230 votes



Source: Commons Votes Services. Excludes 'tellers', the Speaker and deputies

Earnings vary across units even within subjects

Impact on men's earnings relative to the average degree



Source: Institute for Fiscal Studies

{ggplot2} Syntax

Aesthetics

define scale onto which the data are mapped

- color, fill, x- and y-axis
- ``aes()``

Themes

controls the display

- ``theme()``
 - font size
 - axis lines
 - labels and titles

Geometries

the actual visualization form

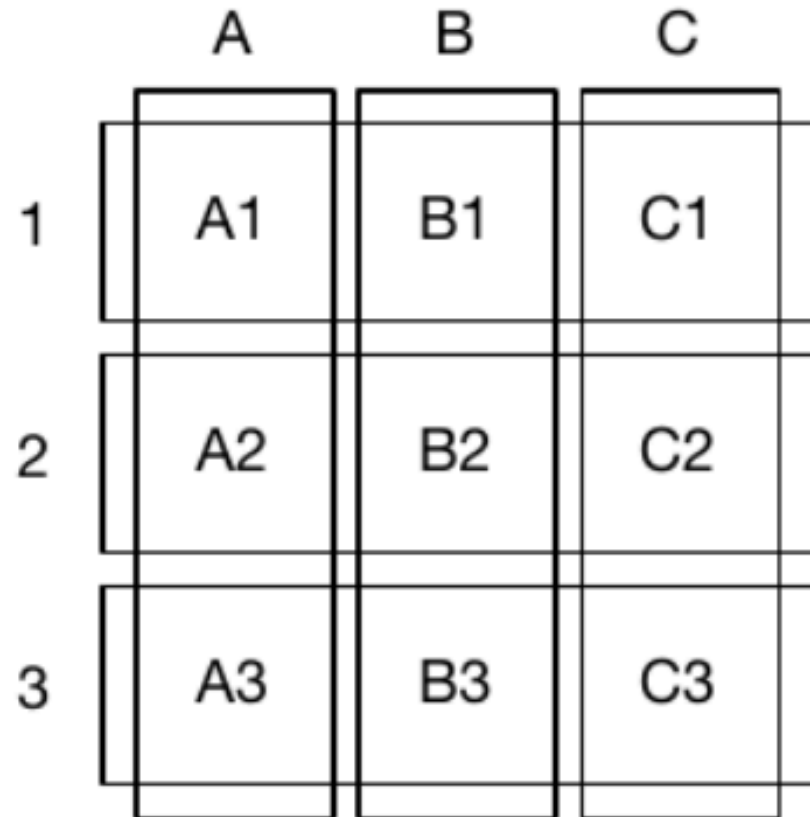
- ``geom_*``
 - `geom_histogram()`
 - `geom_boxplot()`
 - `geom_point()`

Faceting

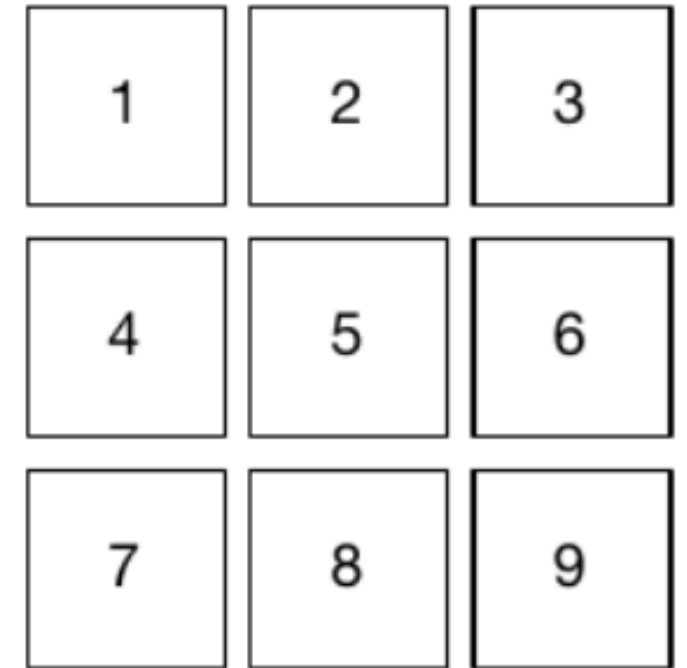
allows for grouping

- ``facet_wrap()``: 1d ribbon panels into 2d
- ``facet_grid()``: 2d grid of panels in rows and columns

**Faceting is
your friend**



facet_grid



facet_wrap

<https://ggplot2-book.org/>

Arranging Multiple Plots

{grid}:
`grid.arrange()`

{ggpubr}:
`ggarrange()`

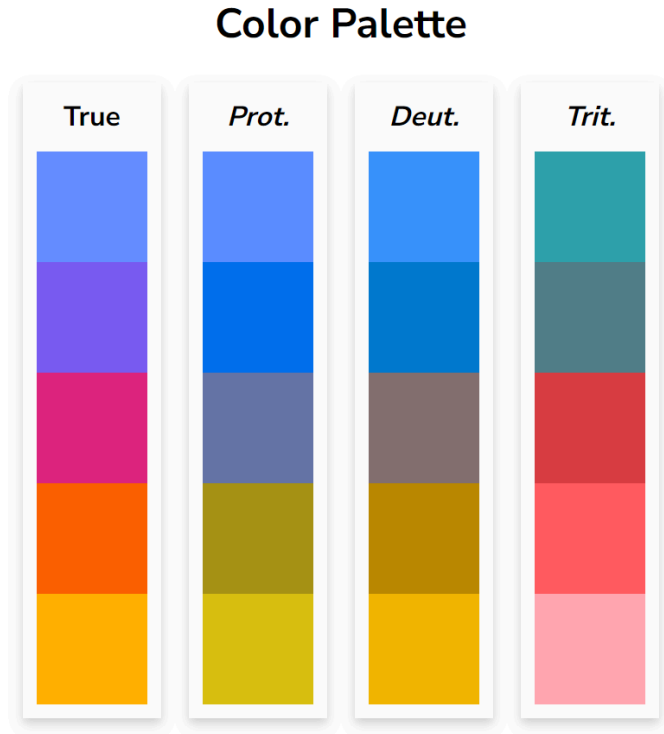
```
grid.arrange(  
  grobs = g1,  
  widths = c(2, 1, 1),  
  layout_matrix = rbind(c(1, 2, NA),  
                        c(3, 3, 4))  
)
```



Illustrating further arguments of `grid.arrange()`, namely `layout_matrix` and relative widths.

Palettes

Color-blindness



Simulator

Palettes

Palettes powered by ML

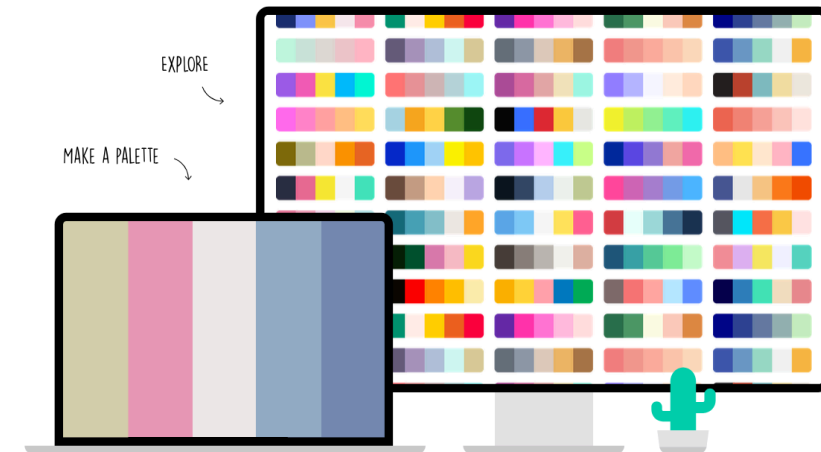
The super fast color palettes generator!

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PRODUCT HUNT
#1 Product of the Month



colors.co

MADE WITH
beautiful.ai